

BiasScope: A Unified Python Library for Bias Detection in Language Models

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Abstract

Language models are now widely used across sectors such as healthcare, finance, and human resources for automated decision-making. However, these decisions can exhibit bias against groups defined by attributes such as gender, race, religion, or age. Measuring this bias is challenging because the relevant metrics are scattered across standalone scripts and benchmark-specific repositories, each with its own input formats, and they span several distinct paradigms, from embedding associations to token probabilities to characteristics of generated text. To address this, we introduce BiasScope, an open-source Python library¹ that standardizes bias evaluation for language models behind a single, consistent interface. BiasScope implements 40 metrics across four families, embedding-based, probability-based, generated-text-based, and prompt-based, so that a model can be assessed across all four paradigms within one framework. Furthermore, the library is designed for easy extensibility, making it possible to incorporate more metrics in the future.

1 Introduction

Language models are increasingly used to make consequential decisions in hiring, healthcare, education, and content moderation. Yet a model can score well on task benchmarks while still assigning higher likelihood to stereotyped sentences (for example, rating “*The nurse finished her shift*” as more probable than “*The nurse finished his shift*”), producing toxic or unequal output for particular demographic groups (for instance, completing prompts that mention Muslims with violent text far more often than prompts about other religions (Abid et al., 2021)), or changing its prediction when a single social-identity term is substituted in the input (Nangia et al., 2020; Nadeem

et al., 2021a; Parrish et al., 2022). Auditing such behavior before deployment has become a fundamental aspect of AI responsibility rather than an afterthought, especially as pre-trained models such as Llama, Gemma, and DeepSeek are trained on web-scale corpora and can inherit and amplify the social biases encoded in that data (Bolukbasi et al., 2016; Caliskan et al., 2017; Bender et al., 2021; Bommasani et al., 2021).

Measuring bias, however, is not a single operation. Bias surfaces at different levels of a model—in its internal representations, in the probabilities it assigns, and in the text it generates—and the field has developed distinct families of metrics to probe each: embedding-based tests that examine geometric associations in representation space (Bolukbasi et al., 2016; Caliskan et al., 2017; May et al., 2019), probability-based tests that compare the likelihoods a model assigns to paired stereotypical and counter-stereotypical sentences (Nangia et al., 2020; Nadeem et al., 2021a), and generation-based analyses that quantify toxicity, regard, and demographic disparities in open-ended output (Gehman et al., 2020a; Dhamala et al., 2021a; Liang et al., 2022). Gallegos et al. (2024) consolidate these into a taxonomy spanning embedding-, probability-, and generated-text-based metrics; we adopt this taxonomy and extend it with a fourth, prompt-based family that evaluates instruction-following and conversational behavior on standardized benchmarks.

The abundance of metrics within each category makes practical application and comparison difficult. Metrics may be distributed as standalone scripts, benchmark-specific repositories, or described in papers without accompanying implementations, and each typically assumes its own input format and preprocessing, which complicates reproducibility. Existing libraries only partially close this gap. LangTest (Nazir et al., 2024) covers stereotype-probability metrics along-

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¹https://github.com/RAINLabLAU/bias_scope

side performance-parity and robustness tests, but omits embedding-association measures and the broader generated-text bias metrics, and it organizes testing by task rather than by the level at which bias arises. AIF360 (Bellamy et al., 2018) and Fairlearn (Bird et al., 2020) provide rich fairness metrics for tabular classifiers but do not support language models. Single-paradigm tools—Perspective API (Lees et al., 2022) for toxicity scoring and BiasAlert (Fan et al., 2024) for generation-level social-bias detection—cover one measurement level in isolation. Consequently, evaluating one model across the full range of bias paradigms in a single, comparable experiment still demands substantial engineering effort.

We introduce BiasScope, a Python library that unifies bias assessment for language models under a single, consistent interface across four complementary families of metrics: embedding-based, probability-based, generated-text-based, and prompt-based. Each family is exposed through shared abstract base classes with consistent validation and a uniform API, so that a model can be assessed under all four paradigms in a single experiment. BiasScope implements a comprehensive suite of previously proposed metrics (Figure 1 and Table 2 in Appendix A): embedding-based measures (WEAT, SEAT, CEAT, and SentenceBiasScore), probability-based measures (CrowS-Pairs, AUL/AULA, CAT/ICAT, and further likelihood-based scores), generated-text analyses (toxicity, regard, counterfactual sentiment, demographic representation, and identity-substitution effects), and prompt-based evaluations built on standardized benchmarks (StereoSet, BBQ, BOLD, RealToxicityPrompts, TruthfulQA). The implementation uses packages such as NumPy (Harris et al., 2020), PyTorch (Paszke et al., 2019), Transformers (Wolf et al., 2020), Datasets (Lhoest et al., 2021), Sentence-Transformers (Reimers and Gurevych, 2019), and LiteLLM.

In summary, this paper makes the following contributions:

1. **A unified, paradigm-complete library** for bias evaluation in language models, spanning embedding-, probability-, generated-text-, and prompt-based metrics—to our knowledge, the first to organize bias assessment by measurement paradigm rather than by benchmark or task. allowing multi-

paradigm evaluation of a single model

2. **A comprehensive and extensible implementation** of 40 bias-evaluation metrics grouped into four families via shared abstract base classes, includes common validation utilities and is easily extendable for adding new metrics.
3. **An open-source release** with documentation and examples, published on PyPI and GitHub under the permissive MIT License to support reproducible bias research and practical model auditing.

2 Related Work

Much progress has been made on bias and fairness in machine learning. Early research developed definitions of fairness and explored how machine learning algorithms could produce discriminatory effects across demographic groups (Hardt et al., 2016). More recent surveys have cataloged types of bias, approaches to fairness, and potential mitigations, giving an extensive account of the challenges in building fair and socially responsible AI (Mehrabi et al., 2021). In natural language processing, a large body of work has shown how social stereotypes based on gender, race, religion, profession, and other demographic traits become encoded in word embeddings and language models, motivating sustained research into bias in language technology (Bolukbasi et al., 2016; Caliskan et al., 2017; Blodgett et al., 2020).

Several open-source toolkits aim to make fairness assessment easier. AI Fairness 360 (AIF360) provides a set of metrics and algorithms to detect bias across different stages of the machine learning pipeline (Bellamy et al., 2018). Fairlearn serves a similar purpose, specializing in fairness evaluation and visualization by comparing model performance across demographic groups (Bird et al., 2020). Other libraries such as Aequitas, FAT Forensics, and Themis-ML support fairness audits, accountability assessments, explanations, and fairness-aware machine learning workflows (Saleiro et al., 2018; Sokol et al., 2022; Bantilan, 2018). These libraries are valuable for conventional machine learning, but they focus primarily on tabular prediction tasks. Closer to the NLP setting, LangTest (Nazir et al., 2024) evaluates language models across robustness, fairness,

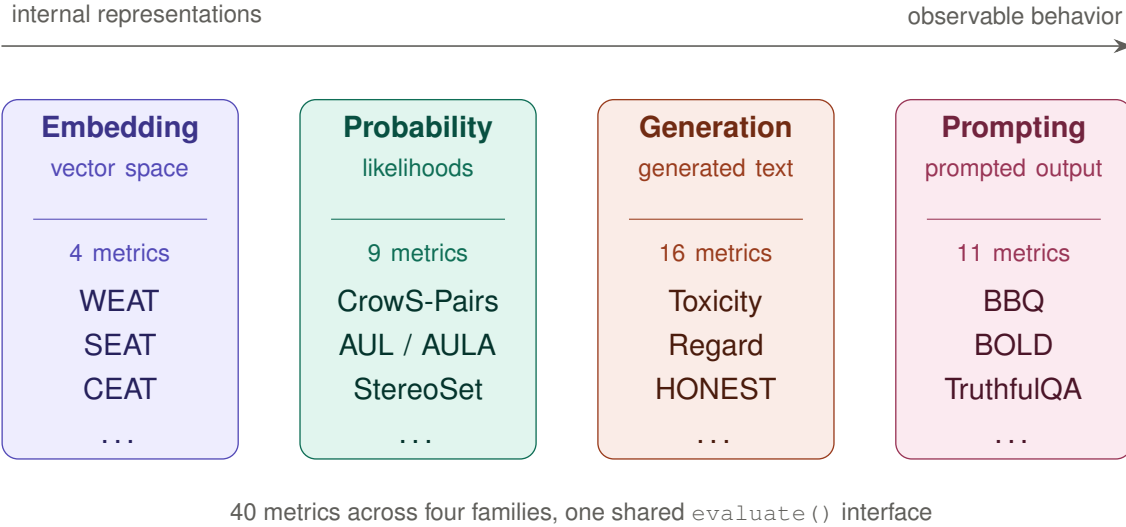


Figure 1: The four bias-measurement paradigms in BiasScope, ordered from internal representations to observable behavior. Representative metrics are shown per family; the full list of 40 metrics appears in Table 2 in Appendix A.

and stereotype tests, but it covers only stereotype-probability metrics, omits embedding-association and broader generated-text bias measures, and organizes testing by task rather than by the level at which bias arises.

Within NLP, a range of metrics and benchmarks measure bias from different angles. Embedding-based measures such as WEAT (Caliskan et al., 2017), SEAT (May et al., 2019), and CEAT (Guo and Caliskan, 2021) quantify associations between demographic categories and target concepts in representation space. Probability-based approaches such as CrowS-Pairs (Nangia et al., 2020), AUL, and AULA (Kaneko and Bollegala, 2022) capture stereotypical preferences from language-model probabilities. Benchmark-driven approaches such as StereoSet (Nadeem et al., 2021a), BBQ (Parish et al., 2022), BOLD (Dhamala et al., 2021a), and RealToxicityPrompts (Gehman et al., 2020a) evaluate stereotype endorsement, social inference, representation, and toxicity in generated text. Generation-based analyses further assess toxicity, regard, demographic representation, and counterfactual sentiment in model outputs (Sheng et al., 2019; Liang et al., 2022). These metrics, however, are typically distributed as standalone repositories or research-oriented software with differing input formats, preprocessing steps, and output formats.

BiasScope builds on these efforts by unifying multiple categories of bias assessment in language models in a single library. Whereas existing fairness toolkits concentrate on predictive-fairness metrics for tabular tasks, and NLP bias measures

are scattered across task-specific repositories, BiasScope organizes embedding-, probability-, generated-text-, and prompt-based metrics behind shared interfaces, validation utilities, and standardized outputs, allowing a model to be assessed across the different paradigms in one framework.

3 BiasScope

BiasScope is a unified library for bias assessment in language models. Rather than treating each evaluation method as a separate tool, it applies every metric through a shared `evaluate()` interface, regardless of whether the metric operates on embeddings, token probabilities, generated text, or prompted model behavior.

The architecture centers on a shared abstract base class, `BiasMetric`, declaring two abstract members: an `evaluate(...)` method and a `category` property. Four intermediate classes inherit from `BiasMetric`, each corresponding to a different family: `EmbeddingMetric`, `ProbabilityMetric`, `GeneratedTextMetric`, and `PromptBasedMetric`. Each of these classes specifies its category and includes input-validation methods required for its specific signal type. Since the `category` property is abstract, every metric must declare its family. This design ensures that all metrics maintain a consistent `evaluate(...)` interface, while the family-specific base classes handle the distinct inputs for each type.

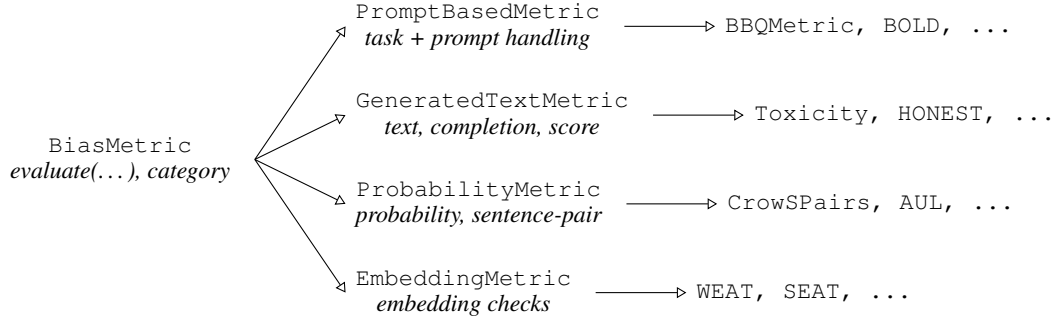


Figure 2: Inheritance structure: concrete metrics (two shown per family) inherit from one of four family base classes, all deriving from the abstract `BiasMetric`.

In addition to the shared interface, the library is designed to support extensibility. Some metrics operate as self-contained benchmark evaluators that load datasets, construct prompts, query a model, and compute aggregate scores. Other metrics are lower-level scoring components that require precomputed embeddings, generated text, lexicons, or user-provided model-scoring functions. This separation between metric logic and model/backend access allows the same evaluation procedures to be reused across different model families and experimental setups. The library also manages optional dependencies through lazy loading. Components that rely on packages such as PyTorch, sentence-transformers, benchmark datasets, Hugging Face models, or external language model interfaces are imported only when needed, keeping the core library lightweight while still supporting more advanced evaluation workflows.

3.1 Metrics

The current version of BiasScope implements 40 bias detection metrics for language models. It is organized into four categories: embedding-based, probability-based, generated-text-based, and prompt-based, as shown in Figure 1.

Embedding-Based Embedding-based metrics measure bias by evaluating relationships between embeddings in a vector space. These metrics examine whether demographic concepts or sentence representations are more strongly associated with particular attribute concepts than expected. This category includes WEAT (Caliskan et al., 2017), SEAT (May et al., 2019), CEAT (Guo and Caliskan, 2021), and SentenceBiasScore (Dolci et al., 2023).

Probability-based Probability-based metrics measure bias by analyzing the probabilities that a language model assigns to individual tokens or alternative versions of a sentence (for example, *The doctor treated his patient* versus *The doctor treated her patient*). Rather than evaluating free-form generated text, these metrics compare the models confidence in stereotypical, anti-stereotypical, or otherwise contrastive alternatives. This category includes CrowS-Pairs (Nangia et al., 2020), CAT (Nadeem et al., 2021b), ICAT (Nadeem et al., 2021b), AUL (Kaneko and Bollegala, 2022), AULA (Kaneko and Bollegala, 2022), LMB (Barikeri et al., 2021), LPBS (Kurita et al., 2019), CBS (Ahn and Oh, 2021), and DisCoMetric (Webster et al., 2020)

Generated-text-based Generated-text-based metrics measure bias in a model’s textual outputs, focusing on properties of the generated text itself, such as toxicity, sentiment, demographic representation, or psycholinguistic characteristics (for example, generating more toxic or more negative continuations for prompts that mention one demographic group than another). This category includes ToxicityFraction (Gehman et al., 2020b), ToxicityProbability (Gehman et al., 2020b), RegardScore (Sheng et al., 2019), ScoreParity (Borkan et al., 2019), SocialGroupSubstitution (Huang et al., 2020), CoOccurrenceBiasScore (Bordia and Bowman, 2019), CounterfactualSentimentBias (Huang et al., 2020), DemographicRepresentation (Liang et al., 2022), StereotypicalAssociations (Liang et al., 2022), MarkedPersons (Cheng et al., 2023), EMT (Gehman et al., 2020a), FGB (Smith et al., 2022), Partial Generation Bias (PGB), HONEST (Nozza et al., 2021), PsycholinguisticNorms (Dhamala et al., 2021a), and GenderPolarity (Dhamala et al.,

2021b).

Prompt-based Prompt-based metrics measure bias from a model’s behavior on benchmark tasks. Unlike generated-text metrics, which score a model’s outputs directly, these define the task and prompts themselves, then query the model and score its responses for properties such as stereotype preference, truthfulness, or counterfactual robustness (for example, testing whether a model’s answer to an ambiguous question shifts with the demographic group it mentions). The category includes StereoSetMetric (Nadeem et al., 2021a), BBQMetric (Parrish et al., 2022), BOLD (Dhamala et al., 2021a), TruthfulQA (Lin et al., 2021), OpinionConsistencyAcrossPersonas, RealToxicityPrompts (Gehman et al., 2020a), DemographicRepresentationBias (Zhao et al., 2018), TofNof (Hong et al., 2025), CounterfactualFairness (Kusner et al., 2017), UnQoverMetric (Li et al., 2020), and AnalogicalReasoningBias.

A description of each metric is provided in Table 2 Appendix A, which summarizes the metrics currently implemented in BiasScope.

3.2 Library Usage

Listing 1 shows how BiasScope evaluates bias with the BBQ prompt-based metric. Line 1 imports the standard `os` module, used to read the API key from the environment. Line 2 imports the `BBQMetric` class from the prompt-based module. Lines 3--6 instantiate the metric with a target model: `model_name` specifies the model in LiteLLM’s `provider/model` format, and `api_key` is read from an environment variable rather than hard-coded, so credentials stay out of the source. Line 7 calls `evaluate`, which loads the BBQ dataset, filters to the ambiguous questions of the Age category, samples `num_samples` of them, queries the model on each, and aggregates the results into a dictionary. In these ambiguous questions the context does not contain enough information to answer, so the correct choice is always the “*Can’t be determined*” option; picking one of the demographic options instead reflects a stereotype. Lines 8--9 print the two headline scores: `bias_score`, the fraction of answers that pick a demographic option rather than “*Can’t be determined*”, and `accuracy`, the fraction that correctly choose it, each formatted as a percentage.

4 Evaluation

We compared the different implemented metrics with those described in their original papers. To evaluate the library, we reproduced one published result from each of the four metric families, using the exact sample sizes and protocols specified in the source papers. The four chosen metrics are: WEAT, CrowS-Paris, MarkedPersons, and BBQ.

4.1 Setup

All experiments were conducted locally on a single NVIDIA RTX A4500 with 20GB of VRAM. All datasets are sourced from public releases.

WEAT (Caliskan et al., 2017) measures whether two target concept sets (e.g., male vs. female first names) associate more strongly with one attribute set (e.g., *career*) than the other (e.g., *family*) in a static word-embedding space, and reports a Cohen’s *d* effect size. We use test 6 from WEAT, which measures the association between gender and career versus family, and scores its 32 stimulus words (16 target names and 16 attribute words) with GloVe 840 B/300 d.

CrowS-Pairs (Nangia et al., 2020) probes stereotype preference in a masked language model by comparing the pseudo-log-likelihoods of minimal stereotype/anti-stereotype sentence pairs and reports the fraction of pairs in which the model prefers the stereotypical side. We use `bert-base-uncased` on all 1 508 pairs from `nyu-ml1/crows-pairs`.

MarkedPersons (Cheng et al., 2023) measures stereotyping in how a language model describes people. When a model is asked to describe a person from a given group, its word choices often differ from group to group. In particular, models tend to describe a *default group* (typically White or male personas) in plain language, while describing other groups with more distinctive, identity-focused words. MarkedPersons finds the words a model uses far more often for a given group than for this default group, since those distinctive words are where stereotypes tend to appear. For example, Cheng et al. (2023) found that descriptions of Black personas repeatedly used words like *resilience* and *strength* that were absent for the default group. To identify such words, the metric applies the Fightin’ Words method (Monroe et al., 2008), which ranks each word by how strongly it is associated with one group over another. In our

```

1 import os
2 from bias_scope.prompts_based import BBQMetric
3
4 metric = BBQMetric(
5     model_name="openrouter/meta-llama/llama-3.1-8b-instruct",
6     api_key=os.environ["OPENROUTER_API_KEY"],
7 )
8
9 result = metric.evaluate(num_samples=20, subset="Age")
10
11 print(f"Bias score: {result['bias_score']:.2%}")
12 print(f"Accuracy: {result['accuracy']:.2%}")

```

Listing 1: Using BiasScope to evaluate bias with the BBQ prompt-based metric.

Family	Metric	Model	Paper	Ours	Rel. Δ
Embedding-based	WEAT-6	GloVe 840B	1.81	1.8139	0.22 %
Probability-based	CrowS-Pairs	BERT-base-uncased	60.5%	58.62%	3.1 %
Generated-text-based	MarkedPersons	gpt-3.5-turbo	8.0396	8.0396	0.0007 %
Prompt-based	BBQ	Llama-3.1-8B-Instruction	0.25	0.2496	0.16 %

Table 1: Reproduction of one representative metric per family. “Paper” is the originally reported value, “Ours” is BiasScope’s output, and “Rel. Δ ” is the relative difference. All four fall within the 10% tolerance.

setup, we do not generate new descriptions. Instead, we reuse the 1,650 persona descriptions that [Cheng et al. \(2023\)](#) published, treating them as a fixed dataset. We apply the metric twice, once using *White* as the default group and once using *male*, and combine the two results as the original paper does.

BBQ ([Parrish et al., 2022](#)) evaluates whether an instruction-following LLM answering a multiple-choice question in an ambiguous context defaults to a stereotype-consistent option rather than the correct “*Can’t be determined*”, reporting a `bias_score` in $[0, 1]$. We evaluate on all 2836 ambiguous items of the `Gender_identity` split of `Elfsong/BBQ` against `Llama-3.1-8B-Instruct`. This is the only experiment where sampling introduces stochasticity (`temperature=1.0`).

4.2 Results

Following established practice in NLP, reproduction is assessed as a graded, quantitative measure of closeness to the original result rather than as exact reproduction ([Belz et al., 2022](#)). We quantify closeness as the relative difference from the reported value and adopt a 10% tolerance as our operational criterion for successful reproduction. We reproduce one representative metric per family as a validation of each category’s implementa-

tion. Table 1 summarizes the reproduction against each paper’s target. All four metrics reproduce within tolerance: relative differences range from 0.0007% (MarkedPersons) to 3.1% (CrowS-Pairs), well inside the 10% band. This indicates that BiasScope’s implementations match the originally reported values.

5 Conclusion

We presented BiasScope, an open-source Python library that unifies bias evaluation for language models across four paradigms: embedding-based, probability-based, generated-text-based, and prompt-based. It exposes 40 metrics through a single `evaluate()` interface with shared validation. Our reproduction study confirms that its implementations match originally reported values within our 10% reproduction criterion across all four families. BiasScope focuses on measurement rather than mitigation; its extensible design makes adding new metrics straightforward. We release it with documentation and examples to support reproducible bias research and practical model auditing.

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- responses) and are exposed through the same `evaluate()` interface.

A Implemented Metrics

Table 2 summarizes the metrics implemented in BiasScope, grouped by metric category. This appendix gives a short definition. Within each family the metrics share an input type (embeddings, model likelihoods, generated text, or prompted

Category	Metric(s)	What it computes
Embedding-based	WEAT, SEAT, CEAT	Measures association patterns between target groups and attribute concepts in embedding space.
Embedding-based	SentenceBiasScore	Measures sentence-level gender bias using a gender direction and importance-weighted word embeddings.
Probability-based	CrowS-Pairs, LPBS	Measures whether a model assigns higher probability to stereotypical than to counter-stereotypical sentence variants.
Probability-based	AUL, AULA	Measures sentence-likelihood bias, with AULA adding attention-based weighting of token contributions.
Probability-based	CAT, ICAT	Measures stereotypical preference in masked-word prediction, with ICAT also accounting for language-modeling quality.
Probability-based	LMB	Measures whether one type of paired sentence is systematically preferred based on perplexity differences.
Probability-based	CBS, DisCoMetric	Measures how demographic attributes change masked-language-model predictions.
Generated-text-based	ToxicityFraction, ToxicityProbability, EMT	Measures different aspects of toxicity in generated text, including prevalence, likelihood, and worst-case tendency.
Generated-text-based	RegardScore, CounterfactualSentimentBias, PsycholinguisticNorms	Measures differences in sentiment or emotional characteristics across demographic groups.
Generated-text-based	ScoreParity	Measures differences in average classifier scores across demographic groups.
Generated-text-based	SocialGroupSubstitution, FGB, PGB	Measures how generated outputs vary across demographic groups or after identity substitutions.
Generated-text-based	CoOccurrenceBiasScore, StereotypicalAssociations	Measures associations between demographic groups and particular attributes or stereotypes in generated text.
Generated-text-based	DemographicRepresentation, HONEST, MarkedPersons	Measures how demographic groups are represented in generated text, including their frequency of mention, association with hurtful terms, or marked lexical descriptions.
Generated-text-based	GenderPolarity	Measures lexical gender polarity in generated text using masculine and feminine term lexicons.
Prompt-based	StereoSetMetric, BBQMetric	Measures stereotype-related bias using standardized benchmark datasets.
Prompt-based	BOLD, RealToxicityPrompts	Measures harmful or biased tendencies in text generated from benchmark prompts.
Prompt-based	TruthfulQA	Measures whether model answers align more closely with truthful references than with common falsehoods.
Prompt-based	OpinionConsistencyAcrossPersonas, TofNof	Measures whether a model changes its stance or behavior across personas or repeated conversational pushback.
Prompt-based	DemographicRepresentationBias, CounterfactualFairness, UnQoverMetric, AnalogicalReasoningBias	Measures demographic sensitivity in model behavior under controlled prompt variations.

Table 2: Summary of BiasScope metrics, organized by category and evaluation target